Slow Momentum with Fast Reversion on a WRDS Futures Panel

Group 67 - Enze Hu (eh3203), Jinrui Hu (jh5260) Completed replication results, November 2, 2026

Executive finding. In the 24-market, fixed-split adaptation, validation-selected CPD improves gross Sharpe from -0.3484 to 1.8803, but falls below long only after 1 bp one-way modeled costs. Enze's complementary five-asset, expanding-window study reports 0.5889 for ordinary LSTM and 0.5182 for selected CPD. Together these results strengthen the motivation to study how market composition, diversification and pooled learning affect CPD's relative performance. They do not establish that adding markets causes Sharpe to rise or makes CPD the best strategy more often.

1. Research question and scope

Wood, Roberts and Zohren (2022) augment a deep momentum LSTM with Gaussian-process changepoint severity and location. We investigate whether those additional features improve a momentum strategy on an alternative WRDS futures panel. Our comparison includes long only, annual and monthly time-series momentum, their equal blend, MACD, the ordinary author LSTM, and five CPD-LSTM variants.

This is a partial method replication. The original paper uses 50 Pinnacle continuous futures, observations from 1990-2020, expanding five-year evaluation windows and a within-asset 90/10 training/validation split. We use 24 provisional distinct underlying markets from WRDS Datastream, train through 2016, validate in 2017-2018 and compare performance in 2019-2020. We do not claim matching original-paper numerical results, a full-calendar tradable backtest, or universal CPD superiority. The project developed through exploratory pilots and is not preregistered. Observations from 2021-2025 remain reserved.

We also incorporate the completed five-asset author-source study committed by Enze Hu on branch Enze-5-assets-reproduction, commit f88874663ba3641f06c867c45446b1c4e5faf93e. Its long-history comparison is complementary evidence, not an additional fold of the 24-market experiment. The original five-asset implementation remains available at that pinned commit; this package includes its aggregate evidence and a reproducible comparison of the committed tables. It does not claim to reproduce that study's private individual-contract data or fitted weights independently.

2. Data and preparation

The source is WRDS LSEG Datastream Futures Continuous Series Data. A 50-series query produced 1,389,945 expanded records, corresponding to 244,960 distinct series/date keys. Repeated keys had identical core prices and relevant metadata; they were collapsed before selecting the coverage-qualified panel. Having multiple contract maturities or duplicated session variants does not create independent underlying assets. The selected codes and provisional market identities are:

Code	Market	Code	Market
2338	CAC 40	2374	Sugar No. 11
2433	US Treasury bond	3801	Full-size S&P 500 index
4122	Euro Bobl	4199	Euro Bund
6409	US Dollar Index	6682	US 5-year Treasury note
8634	Cocoa	9017	US 10-year Treasury note

Code	Market	Code	Market
9020	FTSE 100	9603	STOXX Europe 50
9896	Coffee C	10272	Orange juice
11284	Lumber	12290	EURO STOXX 50
21630	Euro FX	22726	British pound
22970	Class III milk	22990	Japanese yen
23134	Lean hogs	23151	Live cattle
23163	Feeder cattle	23253	Mexican peso

These are provisional underlying matches, not verified exact equivalents of all paper contracts. In particular, code 3801 is not identified as the E-mini S&P 500. Settlement/index values are available, but OHLC, volume and open interest are unpopulated. The continuous-series roll metadata describe a price index and do not establish Pinnacle ratio-adjustment equivalence or a complete executable contract-roll strategy.

The universe was selected using data coverage through the comparison period. This can introduce availability/survivorship selection and is not an ex-ante tradable universe construction.

The immutable selected input snapshot preserves raw price levels. The following explicit preparation policies are applied before feature construction:

- Sugar dates are restricted to dates also observed in both cocoa and coffee. This conservative shared-market restriction is not a fully verified historical exchange calendar.

- Sugar/cocoa/coffee observations on the documented January 3, 2011 special suspension are excluded. Feeder cattle's April 6, 2012 anomalous observation is excluded following settlement-calendar review.

- The Mexican peso scale break on January 13, 2017 starts a new independent segment. No assumed tenfold price correction is applied.

- Gaps exceeding seven calendar days start independent segments; sequences and GP windows cannot cross a break. A remaining raw adjacent index move above 30% stops preparation for review rather than being automatically clipped.

- There is a common 253-observation warm-up so every model can use the full CPD lookback set. Derived author features retain the author's causal EWM winsorization; this does not overwrite source price levels.

There are 87 recorded preparation actions, 37 raw history segments and 55,839 usable feature rows. Ninety-two cutoff/prefix checks confirm that extending the input beyond each cutoff does not change the already-computed features/targets on their common observed prefix. Labels crossing training/validation/test boundaries are excluded from fitting.

The test comparison retains **474 identical signal/return intervals with all 24 assets available**: 11,376 asset observations. The saved inference covers 12,162 available per-asset test endpoints before this common-interval filter. Unmatched market dates are omitted rather than manufactured or forward filled. Consequently the reported annualized statistics are conditional on those retained intervals, using a 252-observation annualization convention. They are not the literal two-calendar-year CAGR or a full daily self-financing portfolio record.

3. Model and selection procedure

We use the author's executable eight-feature implementation: normalized returns over 1/21/63/126/252 observations and MACD pairs (8,24), (16,48), (32,96). This follows the released code where isolated literal numbers in the paper's input-feature prose differ. An LSTM with a time-distributed tanh output generates positions bounded by -1 and 1. Sequence length is **63** for both ordinary and CPD models. A CPD lookback of **252** is a separate GP-history choice; it does not change the LSTM sequence length to 252.

Each CPD variant adds two features, severity and normalized relative location, for one lookback in {10,21,63,126,252}. GPs are fitted on standardized raw-index returns within each independent history. Every window independently initializes the original author's Matérn and changepoint fitting function, with an iteration cap of 200. A failed fit uses only the latest successful result in the same segment, advancing its relative location by the elapsed observations. The saved author-code location feature is relative age, (current index - changepoint index) / lookback. This is the reverse orientation of the location written in the paper's Eq.10; we preserve the executable author convention consistently rather than change fitted inputs after observing results.

The network optimizes the author's Sharpe loss using Adam. Training and validation use the author's nonoverlapping padded batches, with sliding 63-observation inference at test time. There are 523 training sequences and 196 validation sequences, representing 31,809 and 11,844 active labels. Validation aggregates available contributions using fixed N=24, with unavailable assets unallocated. Test returns also use fixed N=24 on retained complete intervals.

Fifty matched hyperparameter candidates are drawn per variant from the author's grid: hidden sizes 5/10/20/40/80/160, dropout 0.1-0.5, learning rates 0.0001/0.001/0.01/0.1, gradient clipping 0.01/1/100 and batches 64/128/256. Maximum epochs are 300, early-stopping patience is 25, and the validation improvement threshold is 1e-4. The search seed is 67; trial seeds are 67 + 1009 times the trial number. This gives 50 ordinary-LSTM fits and 250 CPD-LSTM fits. The overall CPD method therefore has an additional structural window search relative to the ordinary model; the comparison is not a controlled causal attribution of improvement solely to extra features.

CPD lookback	Validation-selected trial	Validation Sharpe	Best epoch	Updates at checkpoint
10	21	1.4610	21	105
21	44	1.4963	36	108
63	38	1.7368	1	5
126	21	1.7430	33	165
252	**22**	**1.8031**	**92**	**460**

The **252** variant is selected by validation before the supplementary fixed-lookback test table is examined. Its settings are hidden size 40, dropout 0.2, clipping 1, learning rate 0.1 and batch size 128. Its trial stopped at epoch 117, preserving epoch 92. The ordinary LSTM selects trial 8, hidden size 80, dropout 0.1, clipping 1, learning rate 0.01, batch size 128 and epoch 22. The best 63-day checkpoint occurs after only one epoch; that is a consequence of validation selection, not evidence that its generalization is reliable.

4. Gross comparison

Asset contributions use the author's 15% annual asset volatility target before equal weighting. This does not imply a 15% portfolio volatility. Raw-index diagnostics are also saved. The following table uses the primary author targets.

Strategy	Conditional annual return	Annual volatility	Sharpe	Max drawdown
Long only	8.1679%	4.7346%	1.6823	8.5421%
Annual TSMOM	-0.2096%	4.5254%	-0.0238	6.1508%
Annual/monthly blend	1.5875%	4.1281%	0.4021	4.7556%
Monthly TSMOM	3.3195%	5.7051%	0.6008	5.5699%
MACD	1.5901%	3.1667%	0.5140	3.5959%
Ordinary LSTM	-0.9713%	2.6970%	-0.3484	7.9485%
LSTM + CPD 10	2.9227%	3.4246%	0.8583	2.5018%
LSTM + CPD 21	-1.2778%	2.4628%	-0.5099	7.1873%
LSTM + CPD 63	2.9537%	3.7778%	0.7894	2.9009%
LSTM + CPD 126	1.6199%	3.7069%	0.4520	4.0589%
LSTM + CPD 252 / validation-selected	**8.5740%**	**4.4274%**	**1.8803**	**2.3843%**

The chosen CPD variant improves the gross risk-adjusted result relative to the ordinary LSTM and long only in this run. It is not a constant-long solution: mean position is 0.2135, mean absolute position is 0.6320 and 65.68% of positions are positive. The negative 21-day result shows that adding CPD does not help uniformly across lookbacks. Unlike the original paper's aggregate result, the longest window is selected here; differences in data, split, universe and sampling prevent a direct numerical comparison. No confidence interval or multi-seed test establishes statistical superiority.

5. Equal-risk presentation

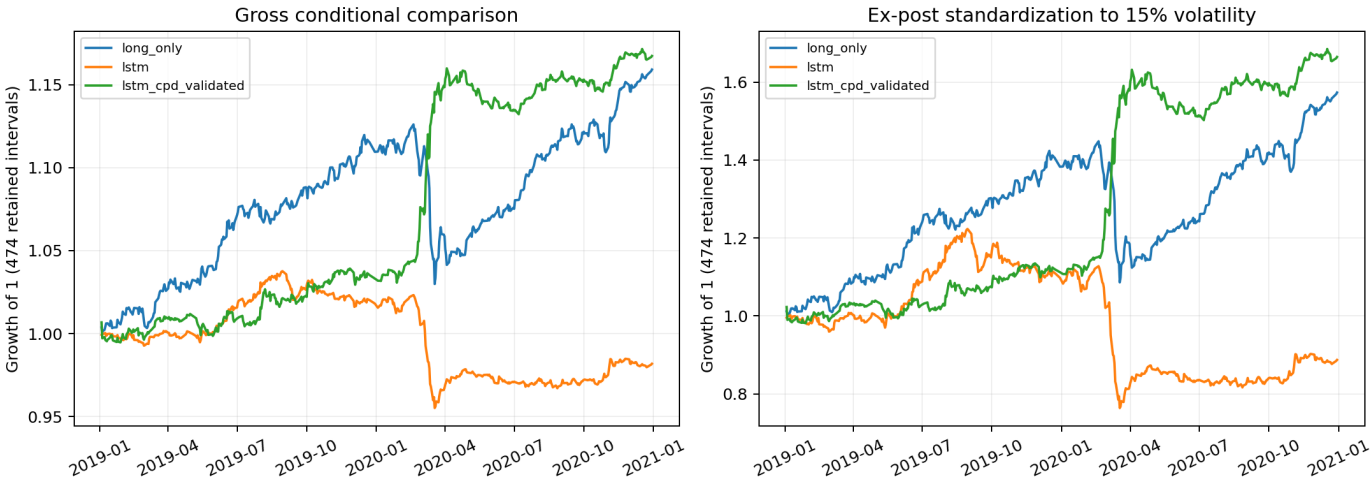
Following the paper's comparison at a common risk scale, each gross portfolio series is multiplied by a single full-common-test constant:

multiplier = 0.15 / (sample standard deviation of retained returns * sqrt(252)).

This is **ex-post standardization**, not a causal leverage strategy. Annualized returns and drawdowns are recomputed after scaling; Sharpe is unchanged.

Strategy	Conditional annual return at 15% vol.	Sharpe	Max drawdown
Long only	27.2401%	1.6823	24.9967%
Ordinary LSTM	-6.1634%	-0.3484	37.5549%
Selected LSTM + CPD	31.1117%	1.8803	7.9318%

All rows have 15% realized sample annualized volatility on the same retained intervals. This standardized return is not an independently observed tradable annual profit forecast.



Gross and risk-standardized conditional results

6. Transaction-cost sensitivity

We apply the paper's Eq.15 using absolute changes in volatility-scaled exposure:

exposure = position 0.15 / (daily_vol sqrt(252)) net return = gross return - cost_bps 1e-4 abs(change in exposure).

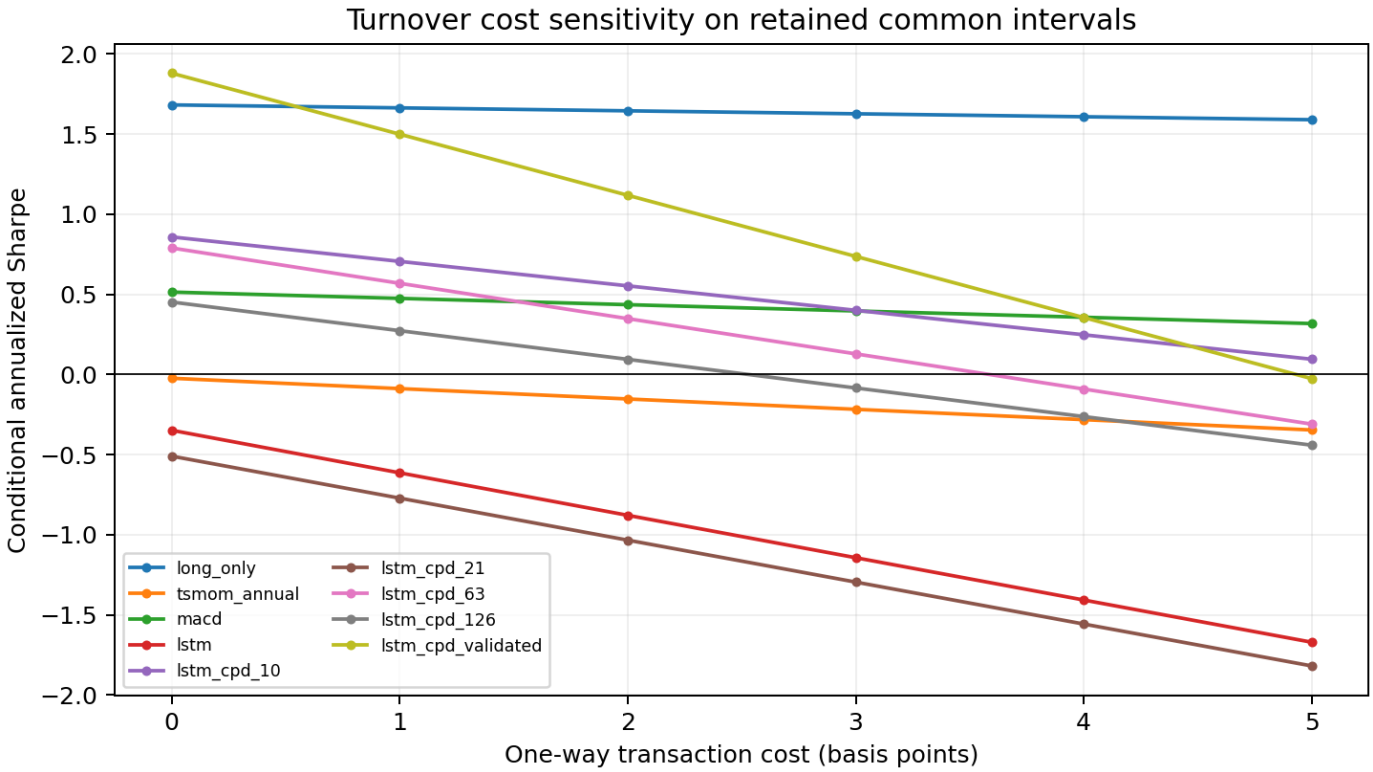
Turnover is computed from all available per-segment test endpoints **before** filtering onto the 474 common intervals. Thus a reversal on a date omitted from the common comparison still changes the immediately preceding position used for turnover at the next retained observation. First available entry turnover is zero, matching the author's implementation; final liquidation is not charged. Costs are applied to original strategy exposure, not the ex-post 15% standardized series.

Strategy	Sharpe, 0 bps	Sharpe, 1 bp	Sharpe, 2 bps	Sharpe, 5 bps
Long only	1.6823	1.6638	1.6453	1.5898
Ordinary LSTM	-0.3484	-0.6140	-0.8792	-1.6716
CPD 10	0.8583	0.7056	0.5529	0.0948
CPD 21	-0.5099	-0.7720	-1.0341	-1.8195
CPD 63	0.7894	0.5687	0.3484	-0.3103
CPD 126	0.4520	0.2731	0.0942	-0.4415
Selected CPD 252	**1.8803**	**1.4995**	**1.1183**	**-0.0264**

The selected model's gross advantage over long only disappears at approximately **0.547 bps one-way**, within this fixed conditional calculation. Its Sharpe crosses zero at approximately 4.931 bps. These are scenario crossover points, not measured market execution costs or universal thresholds. The strongest supported practical finding is substantial turnover sensitivity.

Unmatched dates remain omitted from performance statistics. Costs charged on retained dates do not account for the entire full-calendar portfolio experience. Continuous-index data do not reconstruct all roll trades, discrete contract sizes, bid-ask spreads, funding, margin constraints or exchange-specific fees. The

scenarios therefore do not establish executable net performance.



Conditional turnover-cost sensitivity

7. Verification and reproducibility

All 300 ordinary/CPD trial records are complete and their selected checkpoints agree with validation histories, patience rules and optimizer update counts. The CPD family accounts for 15,582 training epochs. Canonical model/optimizer arrays match the selected checkpoint; original ordinary-LSTM files are preserved.

The full GP computation covers 279,195 windows. There are 154 failed fits (0.0552%): 94/44/10/6/0 for lookbacks 10/21/63/126/252. Every fallback uses the latest past successful fit in the same history, with correctly advanced location. Independent numerical likelihood reconstruction on 465 stored-parameter samples reproduces CPD scores with maximum absolute error 2.7e-10, without refitting GPs. This validates sampled parameter/data consistency, not global optimization.

Uploaded GPU checkpoint replay positions agree with the earlier saved ordinary and selected CPD outputs to about 3e-8. Independent recomputation of all gross, standardized and cost metrics agrees within 1.2e-13; net-return arithmetic within 2e-16. No fresh GPU optimization was performed during the independent audit.

python run_all.py --bundle-dir <private-archive-directory> rebuilds tables, figures and audit results without TensorFlow inference or retraining. The optional --stage full preserves the original scientific scripts and their hashes in an isolated runtime for fresh estimation. README.md, DATA_ACCESS.md, THIRD_PARTY.md and VALIDATION.json distinguish verified reporting from the expensive estimation wrapper that was not rerun during packaging.

8. Differences from the original paper

The following register distinguishes known adaptations, paper/code convention differences, unreplicated analyses and details whose equivalence has not been established. An unverified detail is not treated as a proven methodological match or a proven error in the original study.

Dimension	Original paper or released author implementation	This replication and implication
Data vendor	Pinnacle CLC continuous futures.	WRDS LSEG Datastream continuous price-index series. Matching a model does not make the return construction equivalent.
Number and identity of markets	50 futures spanning commodities, equities, fixed income and FX, with contract/session distinctions in the paper's list.	24 provisional distinct underlying markets. Exact Pinnacle-contract equivalence is not verified; this is not a confirmed literal subset of 24 of the original 50.
Asset mix	The paper includes markets unavailable in our downloaded candidate catalog.	Several energy/metal and other markets are absent; sector composition and correlations change alongside market count. Missing series are not explained by an established licensing cause.
Universe eligibility	Assets enter a test window when sufficient validation history is available; the paper describes less than 10% missing data for selected assets.	Eligibility uses 2017-2018 and 2019-2020 coverage. This can introduce availability/survivorship selection; no equivalent less-than-10%-missing guarantee is claimed.
Historical span	1990-2020, subject to market availability.	Features use 2010-2020. The raw snapshot also contains later observations, but 2021-2025 are excluded and reserved.
Training/validation split	Chronological 90/10 within each asset's expanding estimation history.	Fixed training through 2016 and validation in 2017-2018, after warm-up and segment availability.
Out-of-sample design	Expanding estimation history and successive five-year test windows, with retuning at each expansion.	One 2019-2020 comparison. No five-year expansion or retuning sequence has been reproduced.
Reported aggregation	Main results are averages of the five-year test-window statistics.	Statistics are calculated over one retained interval sequence, rather than averages of the paper's windows. Their numerical levels are not directly comparable.
Calendar and observation coverage	Daily strategy backtests across the paper's evaluation windows. Exact asynchronous-calendar handling is not fully established from the text.	Only 474 identical complete intervals are retained. Omitted dates are not repaired; this is a conditional comparison rather than a complete executable daily portfolio.
Source-specific cleaning	The paper describes causal EWM winsorization; it does not specify the WRDS anomalies encountered here.	Additional shared-ICE-session restriction, specified closure/settlement exclusions, Mexican-peso break and gap resets, plus a review stop for unexplained moves above 30%. These are documented adaptation choices.
Warm-up and history breaks	Sequence availability and original feature construction govern usable observations; an identical universal 253-observation rule is not established in the paper text.	All variants share a 253-observation warm-up. No sequence or GP fit crosses a declared independent segment. This sacrifices some shorter-window observations.
Portfolio denominator	Eq.11 averages contributions over N assets; the eligible universe can differ across expanding windows.	Fixed N=24 on union validation dates, unavailable contributions unallocated; fixed N=24 on balanced common test intervals. Precise equivalence of missing-asset treatment is not claimed.
Input-return/MACD conventions	The paper's input prose prints a 256-observation annual offset and MACD (16,28); the released executable code uses 252 and (16,48).	We follow the released code: 1/21/63/126/252 and (8,24)/(16,48)/(32,96), rather than silently substitute the prose numbers.

Dimension	Original paper or released author implementation	This replication and implication
CPD location convention	Eq.10 describes location increasing from the window's start; the author function returns normalized age from the current observation.	We preserve the executable author feature (t-c)/lookback. This is the opposite orientation of the written location formula, and the trained inputs are not changed after evaluation.
Search controller and stochastic realization	The paper specifies random-grid search and the author repository contains the historical tuner machinery.	A resumable controller uses matched candidates across variants and search seed 67. Exact original sampled candidates/seeds are not reproduced. Final budget remains 50 trials per variant, not the earlier reduced pilot budget.
Repeated-experiment evidence	Section IV-C additionally reports five repeated experiments for the 2015-2020, 21-day-CPD comparison.	One search-seed experiment is reported. We have not reproduced that five-repeat robustness analysis or established confidence intervals.
Software/runtime	Author Keras/TensorFlow and GPflow implementation. Exact historical hardware/runtime equivalence is not established.	Python 3.10, TensorFlow/tf-keras 2.21 and GPflow 2.2.1 in WSL, RTX 5070; two explicit API/dtype compatibility edits. Bitwise equivalence to historical execution is not claimed.
GP execution and fallback diagnostics	Independent per-window author GP fitting; the paper describes fallback to earlier outputs after fitting failure.	The original fitting function and 200-iteration cap are retained, with streaming dispatch and worker recycling for memory. All 154 past-only fills are audited; runtime engineering is changed, not presented as a new GP model.
Portfolio 15% presentation	The paper reports an additional common-risk comparison alongside raw signals.	We apply one ex-post full-common-test multiplier per strategy and recompute metrics. It is not causal risk targeting; exact historical scaling code/window equivalence is not established.
Transaction-cost analysis	Appendix C studies 0-5 bps and averages across its five-year windows.	The same exposure-change formula is applied to our conditional 2019-2020 comparison, computing turnover before the common-date filter. First entry and terminal liquidation charges are omitted; complete roll/contract execution is unavailable.
Economic result	The paper reports broad gross CPD improvements and favorable smaller/intermediate lookbacks, including its validation-optimized strategy.	Validation selects 252 here; the 21-day model loses money. The selected gross winner falls below long only at 1 bp. We do not reproduce the paper's exact rankings, Sharpe levels or cost threshold.
Other original exhibits	The original study includes long-history window comparisons and market/regime case analyses.	We provide gross, standardized and cost comparisons, but have not recreated every original exhibit, historical GP illustration or crisis/Brexit case study.
Development and extension scope	The paper is a separate completed study ending in 2020.	Our work developed through observed pilots. No preregistration is claimed, and no reserved-2021-2025 extension result or causal market-count experiment has yet been produced.

Several central choices are deliberately retained: the author LSTM and Sharpe loss, a 63-observation LSTM sequence, the five CPD lookbacks, the published hyperparameter grid, 50 trials per variant, a 300-epoch cap and patience 25, nonoverlapping training/validation batching, the 15% **asset** risk target, and validation-only checkpoint/window selection. These retained choices should not be described as differences. LSTM sequence length 63 and GP lookback 252 refer to different parts of the model and are compatible with the paper.

9. Exploratory observation: does a larger universe favor CPD?

Our development runs motivate the hypothesis that a larger, diversified market universe **may** improve the absolute Sharpe ratio of LSTM + CPD, its advantage over ordinary LSTM, or the frequency with which it ranks first among strategy benchmarks. These are three distinct outcomes; none has been established as a monotonic or causal function of market count.

Study	Markets	Test scope and aggregation	Ordinary LSTM Sharpe	Selected CPD Sharpe	CPD minus LSTM
Earlier WRDS pilot	1	One 2019-2020 sequence; 505 observations	1.1205	1.0469	-0.0737
Enze author-source study	5	1995-2020; mean of five test-window Sharpes	0.5889	0.5182	-0.0707
Current WRDS panel	24	2019-2020; 474 retained common intervals	-0.3484	1.8803	2.2287

These rows provide cross-study context, not matched estimates of a market-count effect. In particular, an average of window Sharpes is not the Sharpe of a pooled return series, nor is it directly comparable to a single short test period.

9.1 What the five-asset result adds

Enze's study uses S&P 500 day-session futures (SP), 10-year Treasury-note futures (TY), British-pound futures (BN), gold (ZG) and crude oil (ZU). It follows five expanding test windows starting in 1995, 2000, 2005, 2010 and 2015, with a chronological within-instrument 90/10 training/validation split. The final fold includes 2015-2020; the first four folds each span five calendar years. It reports 30 model/window searches and 1,500 candidate fits, using 50 candidates per search.

Five-asset strategy	Mean test-window Sharpe
Long only	0.7679
MACD	0.6492
Ordinary LSTM	0.5889
Validation-selected CPD	0.5182
Fixed 126-day CPD	0.6292

The selected CPD lookbacks are 63, 63, 126, 126 and 21 across the five folds. Selected CPD exceeds ordinary LSTM in three folds, but its mean Sharpe is lower because the losses in the other folds are larger. Fixed 126-day CPD modestly exceeds ordinary LSTM on average; it cannot replace the validation-selected result after inspecting the test set. Neither neural strategy beats long only on the five-asset mean-window comparison. This is evidence of heterogeneous CPD performance, not confirmation of universal CPD improvement.

Its data construction also differs from the 24-market study. Enze compounds same-held-contract returns from WRDS individual-contract histories with explicit roll timing. The 24-market study uses vendor continuous price indices. Gold and crude oil are not in the 24-market pool, so this is not a nested five-market subset that was simply expanded to 24. The environment, search realization, training periods and portfolio/calendar treatment differ as well.

The five-asset means, fold selection and protocol were checked against the committed evidence at the pinned GitHub revision. Original quotes, positions and weights were not available for an independent five-asset raw-return replay. Copies and provenance are in report/evidence/enze_five_assets/; the report pipeline verifies fold averages and exports cross_study_context.csv and five_asset_fold_comparison.csv without refitting models.

9.2 Interpretation of the market-count hypothesis

The pilot values come from paper67_cpd_full_results.zip, specifically cpd_lstm/comparison_2019_2020.csv, rather than a preliminary smaller-budget CPD run. Its source is WRDS code 1825, whose instrument/roll identity remains unverified. It is not a matched one-market subset experiment drawn from the final 24-market panel. Pilot result provenance is saved in report/evidence/pilot_panel_observation.json.

The single- and five-market studies do not show a validation-selected CPD advantage over ordinary LSTM, while the 24-market study does. This additional contrast strengthens the **research motivation** for examining when CPD helps, including a possible role for a broader market universe. It does not strengthen the evidence enough to infer a causal size effect or a higher general probability of being best. Absolute CPD Sharpe does not even rise monotonically across these three reported rows: 1.0469, 0.5182 and 1.8803.

Changing N also changed market composition, source-series identity, data preparation, the evaluation calendar, sampled hyperparameters, selected CPD lookback, training pool size and optimization trajectory. The selected pilot CPD checkpoint used 25 training sequences and one batch per epoch; the panel used 523 sequences and five batches per epoch. Part of the larger CPD-minus-LSTM difference also comes from the ordinary model's deterioration. Neither result isolates asset count. The original paper's 50-market result cannot be added as a third controlled size experiment because its source, periods and evaluation design differ.

There are two plausible mechanisms worth separating. Pooling more histories could help a shared network learn regime behavior. Separately, combining imperfectly correlated strategy returns could lower portfolio risk. Under a stylized equal-volatility, equal-correlation model, Var(R_portfolio) = sigma^2 [1 + (N-1)rho] / N; this can reduce risk when correlations are below one. It does not guarantee higher Sharpe when expected returns, correlations, signal quality and costs also change. Highly correlated or weak-signal additions can provide little benefit or worsen performance. Ordinary LSTM and classical benchmarks can also benefit from diversification, so a rise in CPD's absolute Sharpe would not by itself prove a relative advantage. The working hypothesis is therefore conditional: a broader, sufficiently diverse and consistently constructed universe **may** improve CPD's Sharpe and relative ranking through diversification and more informative pooled training. Testing those mechanisms requires the controlled design below.

In this report, "best" means the highest observed Sharpe among the declared strategy benchmarks for a given equal-weight panel. We did not optimize asset weights or search all asset combinations, so this is not a claim of an optimal asset allocation. The current selected CPD model ranks first **gross**, while long only outranks it at 1 bp; costs must remain part of that definition.

10. Proposed follow-up research - not yet performed

1. **Fix the data and comparison calendar.** Use verified independent underlying markets, consistent continuous-return/roll definitions, a common sample and the same preprocessing. Predeclare choices using training/validation. A formal extension should preserve truly held-out data; 2021-2025 have not yet been evaluated and should not be repeatedly inspected while choosing the design.

2. **Vary size and composition separately.** Within the same 24-market pool, study N in {1,4,8,12,18,24}, with multiple randomly drawn, sector-balanced subsets at each size. For N=24 there is only one distinct full subset, so repeated subset draws do not create new independent portfolios. Expansion beyond 24 requires additional verified markets, not duplicate contracts.

3. **Separate diversification from learning.** First aggregate frozen full-panel predictions over different subsets to examine diversification conditional on the same trained model; this is not an independent smaller-universe training experiment. Then retrain ordinary and CPD models within each subset to test pooled-learning effects. The frozen-model analysis remains descriptive unless carried out under a separately held-out protocol.

4. **Control search and optimization.** Reuse matched candidate grids and multiple seeds, keep all checkpoint/lookback selection on validation, and report equal per-variant search budgets. Because larger panels create more batches per epoch, add a matched-gradient-update comparison rather than confound market count with extra optimizer updates. Also examine comparable compute budgets.

5. **Measure absolute, relative and ranking outcomes.** Report gross and net Sharpe, CPD-minus-LSTM Sharpe, drawdown, turnover and the empirical fraction of predeclared subset/seed/window experiments in which validation-selected CPD beats **all** benchmarks. This win frequency is an empirical statistic, not an established universal probability. Include 0/1/2/5-bp scenarios and equal-risk comparisons without selecting a new winner from the test set.

6. **Quantify uncertainty and failure cases.** Use repeated seeds, independent test windows and paired time-block resampling that preserves cross-asset dependence. Report distributions and uncertainty for both Sharpe and its differences; avoid treating correlated subset draws as independent evidence. Examine whether gains depend on particular sectors, crises or high turnover.

These steps are a proposed research agenda, not completed results. The present evidence supports a gross improvement in one 24-market experiment and a testable universe-size hypothesis. It does **not** establish that adding markets necessarily raises CPD Sharpe or makes it the best strategy more often.

References

Wood, K., Roberts, S., and Zohren, S. (2022). Slow Momentum with Fast Reversion: A Trading Strategy Using Deep Learning and Changepoint Detection. arXiv:2105.13727v3. https://arxiv.org/abs/2105.13727.

Author code: https://github.com/kieranjwood/trading-momentum-transformer, inspected commit e0352cb0bdbf8045accb1bf1a5705ae0cb9ea624.

Calendar/settlement checks used in preparation:

- https://www.ice.com/publicdocs/futures_us/exchange_notices/ExNot2012Holidays.pdf

- https://www.ice.com/publicdocs/futures_us/exchange_notices/ExNot2018HolidayCal.pdf

- https://www.cmegroup.com/tools-information/lookups/advisories/clearing/Chadv12-140.html

- https://www.cftc.gov/sites/default/files/stellent/groups/public/%40rulesandproducts/documents/ifdocs/rul121310iceus006.pdf

Companion five-asset study, Enze Hu, pinned team repository revision:

- https://github.com/EnzeHu0120/ieor4576-slow-momentum-fast-reversion/commit/f88874663ba3641f06c867c45446b1c4e5faf93e

- report/RESULTS.md, report/paper_reference_comparison.csv, report/tables/replication/exhibits_by_fold.csv, report/tables/replication/selected_cpd_windows.csv, and docs/REPLICATION.md at that revision. Aggregate copies and source hashes accompany this submission.